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Fairness-aware PageRank via Edge Reweighting



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Confirmation: I certify that my OpenReview profile is up to date (including accurate first and last names, a valid preferred email, and current and past affiliations) and that this paper adheres to the guidelines in the Call for Papers, including policy on human participants, authorship, and limits on maximum authorship.

Abstract:
Link-analysis algorithms, such as PageRank, are instrumental in understanding the structural dynamics of networks by evaluating the importance of individual vertices based on their connectivity. Recently, with the rising importance of responsible AI, the question of fairness in link-analysis algorithms has gained traction.

In this paper, we present a new approach for incorporating group fairness into the PageRank algorithm by reweighting the transition probabilities in the underlying transition matrix. We formulate the problem of achieving fair PageRank by seeking to minimize the fairness loss, which is the difference between the original group-wise PageRank distribution and a target PageRank distribution. We further define a group-adapted fairness notion, which accounts for group homophily by considering random walks with group-biased restart for each group. Since the fairness loss is non-convex, we propose an efficient projected gradient-descent method for computing locally-optimal edge weights. Unlike earlier approaches, we do not recommend adding new edges to the network, nor do we adjust the restart vector. Instead, we keep the topology of the underlying network unchanged and only modify the relative importance of existing edges. We empirically compare our approach with state-of-the-art baselines and demonstrate the efficacy of our method, where very small changes in the transition matrix lead to significant improvement in the fairness of the PageRank algorithm.

Submission Number: 2146

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Paper Decision

Decision by Program Chairs  20 Jan 2025, 13:24 (modified: 23 Jan 2025, 09:50)

 Program Chairs, Authors  Revisions (/revisions?id=D8nHhS9Yi1)

Decision: Reject

Comment:

Paper metareview: The paper proposes a fairness-aware modification of the PageRank algorithm by reweighting transition probabilities to achieve group fairness, using a projected gradient descent method for optimization. While the paper introduces an interesting concept of group-adapted fairness and demonstrates some empirical results, several critical issues undermine its contribution. Theoretical analysis, such as convergence guarantees and complexity analysis of the proposed method, is lacking, raising concerns about the validity and scalability of the approach. The experimental evaluation does not convincingly demonstrate superiority over existing baselines and focuses primarily on fairness loss and transition matrix changes without providing clear metrics for evaluating the quality of the adjusted PageRank vector. Additionally, key aspects of the methodology, such as the grouping of nodes and the practical implications of fairness modifications, are insufficiently explored. Overall, while the idea has potential, the paper requires significant improvements in theoretical rigor, clarity, and comprehensive evaluation to justify its acceptance.

Official Review of Submission2146 by Reviewer 8Rry

Official Review by Reviewer 8Rry  03 Dec 2024, 10:50 (modified: 03 Dec 2024, 14:29)

 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Reviewer 8Rry, Authors

 Revisions (/revisions?id=kt5WBxj1ZM)

Review:

This paper studies fairness-aware PageRank by reweighting the transition probabilities. The key idea is to find the (restricted) group-adaptive ϕ -PageRank vector by applying projected gradient descent.

Strengths

Overall, the proposed FairGD achieves a smaller fairness loss with a smaller change in transition matrix.

Weaknesses

W1. When introducing group-adapted PageRank, the authors claim that "the total PageRank scores in group V_k are largely contributed by random walks (re)starting at vertices in group V_k ." Could the authors explain why this is true, since PageRank should be the stationary distribution? It is a bit unclear why restarting at specific vertices would contribute to a stationary distribution.

W2. There are four problem definitions in Section 3, and it is unclear which problem the author is trying to solve. Is it problem 4?

W3. It would be better if the authors could explain what the advantages/benefits of group-adapted fairness loss over the original ϕ -fairness loss in fairness-aware PageRank.

W4. In Figure 2, why does FairGD have a higher transition matrix change on Book when ϕ is large and Twitter when ϕ is small?

W5. The experiments seem mostly focus on the change of transition matrix and fairness loss. It doesn't evaluate how good the fair PageRank vector is after adjustment (the only one seems to be a visualization in Figure 5). The author might want to explain what the effectiveness metric is for this graph ranking problem. Can we have a quantitative measure to evaluate Figure 5?

W6. The paper might be further improved if there can any theoretical analysis on the upper bound of change to the ranking vector by FairGD.

Questions:

Please see review above.

Ethics Review Flag: No

Scope: 4: The work is relevant to the Web and to the track, and is of broad interest to the community

Novelty: 4**Technical Quality:** 4**Reviewer Confidence:** 3: The reviewer is confident but not certain that the evaluation is correct

Rebuttal by Authors

Rebuttal

by Authors (Haoyun Zhou (/profile?id=~Haoyun_Zhou1), Honglian Wang (/profile?id=~Honglian_Wang1), Aristides Gionis (/profile?id=~Aristides_Gionis1))

05 Dec 2024, 20:24

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors

Rebuttal:

Thank you for your questions. Please find our responses below.

W1

In homophilic graphs, a random walk starting from a red node is more likely to stay in the red group than the blue group.

Figures 3 and 4 in reference [30] demonstrate that red nodes in homophilic graphs primarily retain their personal PageRank scores within the red group. If random walk is forced to restart uniformly from the red group, the final PageRank score, which is calculated as the average of all personalized PageRank scores of red nodes, will disproportionately favor the red group over the blue group.

W2

We studied all four questions. Algorithm 1 is designed for problems 1 and 2, and algorithm 4 is designed for problems 3 and 4.

W3

The original ϕ -fairness loss involves optimizing the jump vector, while the group-adapted fairness loss is defined over a set of meaningful jump vectors, $v_1, \dots, v_l$. We note that these jump vectors are part of the problem's setup, not the solution itself. The group-adapted fairness loss focuses on optimizing edge weights, which we believe is more actionable and feasible than optimizing the jump vector.

W4

For the book dataset, the original and target group PageRank scores are $(0.10, 0.48, 0.42)$ and $(\phi, 1 - \frac{\phi}{2}, 1 - \frac{\phi}{2})$, respectively. When ϕ is small, the two distribution are similar, leading to small fairness loss and weight changes.

For the Twitter dataset, the original and target group PageRank scores are $(0.42, 0.58)$ and $(\phi, 1 - \phi)$. As ϕ moves towards 1, the target diverges from the original PageRank score, thus both fairness loss and weight change increases.

W5

Our definition of good adjustment is to minimize changes to the original transition matrix instead of to limit changes to the PageRank vector for two reasons. First, the original PageRank vector may itself be unfair, thus staying close to the original PageRank score could be undesirable. Second, introducing a constraint to limit changes to the PageRank vector would over-constrain the problem, and complicate its resolution.

W6

Assume there are l groups, and the proportion of nodes in each group is represented as $(P_1, \dots, P_l)$. Since the lower bound PageRank score for group j is $P_j \cdot \gamma$, the upper bound for group j can be expressed as $1 - (1 - P_j)\gamma$.



➔ *Replying to Rebuttal by Authors*

Official Comment by

Reviewer 8Rry

Official Comment by Reviewer 8Rry 📅 09 Dec 2024, 05:20

👁 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors

Comment:

I appreciate authors' efforts to address my concerns. However, several of them are still not addressed.

W1. I understand what you mean by '(re)starting' here now. I think the description is a bit confusing as 'restarting' gives me an impression that it is talking about restarting in one round of random walk. But the 'restarting' in the paper is referring to the jump (teleportation) vector. A better description would improve the clarity.

W2. Thank you for the clarification. It would be better if the authors could reflect it explicitly in the paper.

W3. Could you explain more on why optimizing edges weights is more feasible and actionable? And under which condition (or real-world application) we would have a set of jump vectors to choose from? Both transition matrix and jump vector are inputs to PageRank, so I think optimizing transition matrix/jump vector is just alternative to each other.

W4. I think it is still unclear why it only happens to these two datasets. What about the original and target PageRank for other datasets?

W5. I agree that original PageRank might be unfair. But it is still important to understand the quality of ranking, as a ranking could be absolutely fair but meaningless.

W6. Thank you for the insights. Could you provide more details on how you derive the lower bound and upper bound?

➡ *Replying to Official Comment by Reviewer 8Rry***Official
Comment by
Authors**

Official Comment

by Authors (👁 Haoyun Zhou (/profile?id=~Haoyun_Zhou1), Honglian Wang (/profile?id=~Honglian_Wang1), Aristides Gionis (/profile?id=~Aristides_Gionis1))

📅 11 Dec 2024, 07:42

👁 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors

Comment:

W1: Thank you for your comment! In the literature, the terms "restart vector" and "jump vector" are often used interchangeably. By "restarting at vertices in group V_k " we mean setting the jump (restart) vector v such that $v[j] = \frac{1}{V_k}$ if node j belongs to group k , and $v[j] = 0$ otherwise. We will make sure this is clarified in the revised version of the paper.

W2: Thank you for your comment. We will ensure that this is clarified and made explicit in the revised version of the paper.

W3: Although both the transition matrix and the jump vector are inputs to the PageRank algorithm, the jump vector is more of a theoretical construct, representing probabilities that are largely beyond our control.

Consider a video recommendation graph as an example. Here, each node represents a video, and an edge (u, v) indicates that while a user is watching video u , video v is recommended as the next video to watch. Typically, there is a list of recommended next-to-watch videos, and the ranking of each video can be influenced by the edge weights. For instance, a recommendation list $[v_1, v_2, \dots, v_k]$ corresponds to edge weights $w(u, v_1) \geq w(u, v_2), \dots, \geq w(u, v_k)$. In this scenario, a moderator can adjust the rankings of the recommended videos, which in the graph translates to modifying the edge weights.

The situation is different for the jump vector. In this context, the jump vector could represent the probability distribution of where a user might start if they either (1) ignore the recommendation list or (2) choose a video to watch on their own. This distribution reflects the user's intrinsic preferences and is not something that can be directly influenced or controlled.

W4: Thank you for your question, we will provide a more detailed analysis regarding the transition matrix change in the next version of the paper.

The original group-wise pagerank scores are listed in the last column of Table 1. The target PageRank scores are calculated by setting the first group's score to $\phi_1 = \phi$, and distributing the remaining PageRank score, $1 - \phi_1$, equally among the other groups, as explained in lines 673-678.

For the Blogs dataset, the scores (0.48, 0.52) are nearly evenly distributed between the two groups. Consequently, changes to the transition matrix are substantial for both small and large ϕ values. In the Mind dataset, the original scores are (0.26, 0.74), though they were incorrectly listed as (0.74, 0.26) in the paper. As ϕ increases, the target scores diverge more significantly, resulting in larger weight changes. For the Slashdot dataset, the initial scores are distributed almost evenly across four groups (0.3, 0.17, 0.36, 0.17). The target scores are $(\phi, \frac{1}{3}(1 - \phi), \frac{1}{3}(1 - \phi), \frac{1}{3}(1 - \phi))$. As ϕ approaches 0.9, the target scores deviate significantly from the original scores, leading to larger weight changes for higher ϕ values.

W5: We believe that the meaningfulness of the ranking is ensured by remaining faithful to the original transition matrix. This fidelity is crucial for maintaining the problem's practical relevance, as a moderator's ability to manipulate edge weights is inherently limited.

W6: Assume there are l groups, let $[l] = 1, \dots, l$ and let the proportion of nodes in each group be denoted as $(P_1, \dots, P_l)$. Let U_j denote the group-wise PageRank score of group j , for all $j \in [l]$.

First we have $1_k^T p = (1 - \gamma)1_k^T P^T p + \gamma 1_k^T v \geq \gamma 1_k^T v$ (this is also stated in lines 291-294). For group k , we have $U_k = 1_k^T p \geq \gamma 1_k^T v = \gamma P_k$, where the last equality holds because v is a uniform vector. Thus we proved the lower bound of U_j is γP_k .

It follows that $U_k = 1 - \sum_{j \in [l] \setminus k} U_j \leq 1 - \sum_{j \in [l] \setminus k} \gamma P_j = 1 - \gamma \sum_{j \in [l] \setminus k} P_j = 1 - \gamma(1 - P_k)$. In other words, the upper bound of U_k is $1 - \gamma(1 - P_k)$.

We will ensure this is explained in detail in the next version of the paper.

We thank the reviewer again for the valuable comments and discussion!

Official Review of Submission2146 by Reviewer 8pxn

Official Review by Reviewer 8pxn 📅 02 Dec 2024, 16:49 (modified: 03 Dec 2024, 14:29)

👁 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Reviewer 8pxn, Authors

📄 Revisions (/revisions?id=LpyUPAw7br)

Review:

This paper proposes to incorporate group fairness into PageRank by re-weighting transition probabilities in the underlying transition matrix and then propose a gradient-descent method to compute locally-optimal edge weights. The authors conduct experiments on five datasets and compare their results to four baselines. Experimental results demonstrate their methods' efficacy.

Pros: 1. This paper introduces the idea of "group-adapted fair PageRank," which may be useful for the analysis of fairness-aware PageRank. 2. The experimental results seem to be good. 3. The writing is good.

Cons: 1. It is unclear how the algorithm groups nodes and how the number of groups is set. 2. The experimental results in Section 5.4 do not demonstrate the superiority to LFPR [30].

Questions:

1. How do you define the group of nodes? How many groups are set for each dataset in your experiments?
2. What if each node is treated as a group and all nodes are grouped into a group? Have you considered the two extreme situations?

Ethics Review Flag: No**Scope:** 3: The work is somewhat relevant to the Web and to the track, and is of narrow interest to a sub-community**Novelty:** 3**Technical Quality:** 4**Reviewer Confidence:** 3: The reviewer is confident but not certain that the evaluation is correct

Rebuttal by Authors

Rebuttal

by Authors (Haoyun Zhou (/profile?id=~Haoyun_Zhou1), Honglian Wang (/profile?id=~Honglian_Wang1), Aristides Gionis (/profile?id=~Aristides_Gionis1))

05 Dec 2024, 20:26

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors

Rebuttal:

Thank you for your questions and comments. Please find our responses below.

Q1

For the datasets Books, Blogs, MIND, and Twitter, we use node labels as group indicators. For the Slashdot dataset, since node labels are unavailable, we apply the METIS graph partitioning algorithm (referenced in [13] of the submitted paper) to divide the vertices into groups.

The last column of Table 1 indicates the number of groups, corresponding to the size of the original group-wise PageRank score tuple. We will clarify this in the next version of the paper.

Q2

Our framework can accommodate both extreme cases:

(1) Each node is treated as a group. In this scenario, we can modify Equation (3.1) by setting the group-wise target PageRank scores, ϕ , as a n -dimensional vector, where ϕ_k represents the target PageRank score for node k . Additionally, 1_k can be set as the indicator for node k . In Equation (3.4), we can further set v_k to be the indicator of node k . (2) When all nodes belong to one group, there are no fairness issues to address. If we need to assign specific PageRank score to certain subgroup, we can modify ϕ , 1_k , and v_k in Equation (3.1) and (3.4) accordingly.

cons (The experimental results in Section 5.4 do not demonstrate the superiority to LFPR [30])

We compared FairGD with LFPR on the Blogs and Mind dataset. On the Mind dataset, FairGD achieved both smaller fairness loss and smaller weight changes compared to LFPR. On the Blogs dataset, although FairGD exhibited slightly higher fairness loss than LFPR, the absolute fairness loss was minimal (less than 0.015), indicating that both algorithms produced near-perfectly fair solutions. Furthermore, FairGD outperformed LFPR in weight change, and achieved smaller weight changes for 8 out of 9 ϕ values.

These results confirm that FairGD effectively achieves fairness with minimal modifications to the transition matrix, showcasing its strengths relative to LFPR.

Official Review of Submission2146 by Reviewer uQRE

Official Review by Reviewer uQRE 02 Dec 2024, 07:42 (modified: 03 Dec 2024, 14:29)

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Reviewer uQRE, Authors

Revisions (/revisions?id=9H2YeU7D2m)

Review:

This paper proposes a new method to incorporate fairness into the PageRank algorithm by reweighting edges, aiming to achieve globally fair PageRank without requiring strict local fairness at each vertex, and reducing modifications to the transition matrix. The method introduces a new concept of group-adapted fairness, taking into account the random walks of each group and group-biased restarts. Finally, an efficient projected gradient descent algorithm is proposed to find the locally optimal edge weights, and it is compared with four baselines on five datasets, showing promising results.

Strengths:

- 1. The suggested approach aims to integrate group fairness into the PageRank algorithm by adjusting the transition probabilities within the transition matrix. This modification demonstrates good improvements in aligning with diverse fairness objectives, outperforming the previous baseline methods.
- 2. The paper is generally structured clearly and well-written. The experimental design is comprehensive, with a rich set of given datasets and baseline methods. Additionally, the study offers an exploration of the trade-off between fairness loss and weight adjustments during the experimentation, providing valuable insights into the balance between these factors.

Weaknesses:

- 1. While the primary focus of the article is on inter-group fairness, there appears to be limited exploration regarding the potential impact on intra-group individual fairness.
- 2. The analysis of the experimental outcomes lacks the comprehensiveness necessary for a robust evaluation. Furthermore, the paper fails to provide a satisfactory explanation for the less-than-ideal performance of the FairGD method when applied to the Blogs dataset, leaving the underlying causes for this discrepancy unexplored.

Questions:

- 1. The grouping of vertices in the paper is quite clear, but have you taken into account more intricate scenarios, such as those involving nested groupings?

Ethics Review Flag: No

Scope: 3: The work is somewhat relevant to the Web and to the track, and is of narrow interest to a sub-community

Novelty: 5

Technical Quality: 5

Reviewer Confidence: 3: The reviewer is confident but not certain that the evaluation is correct



Rebuttal by Authors

Rebuttal
by Authors (Haoyun Zhou (/profile?id=~Haoyun_Zhou1), Honglian Wang (/profile?id=~Honglian_Wang1), Aristides Gionis (/profile?id=~Aristides_Gionis1))
 05 Dec 2024, 20:27 (modified: 05 Dec 2024, 20:32)
 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors
 Revisions (/revisions?id=ZAB2aZfrLn)

Rebuttal:

Thank you for your questions and comments. Please find our responses below.

Weaknesses 1

While we did not explicitly study the intra-group individual fairness, our framework can accomodate it. For instance, if we aim to assign the same PageRank score to all nodes within a group, we can define an individual target vector ϕ_I , where $\phi_I[j]$ represents the target PageRank score for node j . We can modify Equation (3.1) into $L(P, \gamma, v) = \frac{1}{K} \sum_{k=1}^K (1_k^T p - \phi_k) + \sum_{j=1}^b (p[j] - \phi_I[j])$, where p is the PageRank vector corresponding to parameters (P, γ, v) . This formulation incorporates individual fairness as an additional term in the fairness loss function, and enables the study of both group and individual fairness within the same framework. We believe that this is a nice idea and we will investigate it further in our followup work.

Weaknesses 2

Please refer our answer to **cons** to reviewer 8pxn.

Q1

Although we did not address this scenario in the draft, our framework is capable of handling such cases. Suppose the nested grouping structure can be represented as a tree, where the root node denotes all vertices belonging to a single group, and the leaves indicate each vertex as an individual group. Each layer of the tree represents a different grouping of vertices.

To account for this nested grouping structure, we can compute a loss function for each layer (i.e., each grouping of vertices) based on the definitions in Equations (3.1) and (3.4). The problem can then be approached by minimizing the sum of the loss functions across all layers.

Official Review of Submission2146 by Reviewer Xfax

Official Review by Reviewer Xfax 📅 20 Nov 2024, 18:32 (modified: 03 Dec 2024, 14:29)

👁 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Reviewer Xfax, Authors

📄 Revisions (/revisions?id=8hUBrGz971)

Review:

The authors propose a pair of new fairness inducing formulations for PageRank by modifying the probability transition matrix of the random walk to minimise group-fairness violations via edge re-weighting. Since the problems are non-convex, projected gradient algorithms are used to obtain approximate solutions. While the approach is reasonable, I have some concerns regarding the technical merit.

(1) No convergence analysis of FairGD or its variants are provided. Hence, it is not clear if the algorithm converges to a stationary point of the non-convex fairness formulation, even if the full gradient (without approximation) is used. In practice, the authors employ an approximate gradient, and its not clear in what way this step affects the convergence.

(2) Related to the above observation, I am not sure if the projection step is correct for the restricted-PR problem. The constraint set of problem (3.3) is convex, and it is not clear whether steps 3-9 of Algorithm 3 are indeed the valid steps. Unless the authors analyse the KKT conditions of the projection operation, derive the requisite conditions for optimality, and establish the correctness of Algorithm 3, it is hard to say that this is a principled way of implementing projections.

(3) A complexity analysis of FairGD and its variants is lacking. Scalability seems to be an issue in general since the number of variables scales quadratically with the number of vertices of the graph. This raises the concern that the algorithm can only be applied to small graphs.

(4) Although the proposed formulation does not add extra edges to the graph, I don't see any reason why it cannot delete pre-existing edges from the graph by assigning a value of zero to a particular entry of the transition matrix. This can occur for example, from Step 2 of the gradient projection operation. In that sense, the algorithm can directly modify the topology of the network.

(5) For a large number of groups K , the proposed loss function may not be a meaningful indicator of fairness violations. This is because the authors consider the sum of the violations across groups. Hence, it is possible that the computed solution induces a small violation for some groups, at the expense of other groups. Perhaps a better performance indicator would be to not to report the average loss, but the worst loss over all groups.

Questions:

(1) Is it possible to establish convergence of the projected gradient algorithm to stationary points of the non-convex problem?

(2) Please verify if Algorithm 3 is indeed the correct projection onto the set of constraints of Problem (3.3)

(3) Please specify the complexity of the algorithms.

(4) Since edge deletions are permitted, can it result in a set of vertices getting disconnected from the rest of the graph? This is not captured by the current metrics for measuring changes in the transition matrix.

(5) Is there an intuitive/theoretical explanation for why the fairness losses (depicted in Figure 1a and 3a) behave the way they do as a function of ϕ ?

(6) If LFPR_n and LFPR_u are both guaranteed to achieve ϕ -fairness, why don't they achieve zero fairness loss?

Ethics Review Flag: No

Scope: 4: The work is relevant to the Web and to the track, and is of broad interest to the community

Novelty: 3

Technical Quality: 3

Reviewer Confidence: 3: The reviewer is confident but not certain that the evaluation is correct



Rebuttal by Authors

Rebuttal

by Authors (👁 Haoyun Zhou (/profile?id=~Haoyun_Zhou1), Honglian Wang (/profile?id=~Honglian_Wang1), Aristides Gionis (/profile?id=~Aristides_Gionis1))

📅 05 Dec 2024, 20:31

👁 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors

Rebuttal:

Thank you for your questions; we will clarify the time complexity and provide formal convergence analysis in the next version.

Q2:

Our problem constraint involves the intersection of two convex regions. Specifically, the constraint $(1 - \delta)P[i, j] - \epsilon \leq \hat{P} \leq (1 + \delta)P[i, j] + \epsilon$ defines a box, and the constraint $\sum_{j \in [n]} P[i, j] = 1$ defines a simplex.

In Algorithm 3, Lines 5–6 project a candidate solution onto the box, specifically, it implements

$$P_{\text{box}}(\mathbf{x})_i = \begin{cases} l_i & \text{if } x_i < l_i, \\ x_i & \text{if } l_i \leq x_i \leq u_i, \\ u_i & \text{if } x_i > u_i. \end{cases}$$

and Lines 7–8 employ the widely applied normalization technique to map the projected point in the box to a feasible solution.

Q1:

With a sufficiently large t , the approximated gradient y_k will become tight and eventually achieves machine precision. When the gradient is exact, the smoothness of the objective function guarantees convergence to a local minimum. Furthermore, the feasibility of the updated transition matrix ensures that the KKT conditions are satisfied.

Q3:

The primary computational load of the FairGD algorithm lies in line 8, where matrix multiplication is performed for t times, As a result, each gradient descent step has a time complexity of $O(n^2t)$.

Q4:

Indeed, the constraint $C(R)$ allows edge deletion, which is why we also propose an alternative constraint, $C_R(P)$. By setting small values for δ and ϵ , we can prevent edge deletions from the graph, and ensure its structure remains intact.

Q5:

Indeed, the loss function could be formulated to minimize the maximum loss among all groups, this can be a very interesting future work to explore!



➡ *Replying to Rebuttal by Authors*

Response to rebuttal

Official Comment by Reviewer Xfax 📅 09 Dec 2024, 11:39

👁 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors

Comment:

I thank the reviewers for their responses. However, some of the responses regarding the technical details are still unsatisfactory.

Q2: I am still not convinced that lines 5-8 actually implement the Euclidean Projection onto the intersection of the two convex sets. I assume that this is what the authors are trying to compute, since Algorithm 3 is currently titled "Projection to feasible set". Since the described procedure first projects onto the box constraints, followed by the simplex constraints, it is reminiscent of alternating projections on convex sets, but it is difficult to be sure because the authors don't present it as such. Second, even if the above interpretation of Algorithm 3 is correct, lines 7-8 do not resemble the Euclidean projection onto the simplex (see Section 6.2.5, <http://dx.doi.org/10.1561/24000000003>), i.e., it is not row normalization. This would be the correct step if projections onto the simplex were performed using KL-divergence, and not the euclidean distance metric. Hence, it seems that the authors have used an alternating combination of two different projection schemes, and stated without proof that this is the optimal projection. This is not obvious at all from the present description of Algorithm 3, and needs more rigorous technical arguments to be substantiated.

Q1: Lets assume that exact gradients are used. Even in this scenario, the proof sketch provided by the authors is not sufficient, because **(A):** a step size schedule needs to be specified to guarantee sufficient descent at each iteration and **(B):** the projection step should produce iterates that optimally solve the projection problem (mere feasibility is not enough). Regarding the first point, the authors have not provided any details, and regarding the second, please see my previous comment. Hence, it is not obvious that convergence to a KKT point can be reached as the authors have claimed.

Q3: Thank you for the clarification. It seems that there is a trade-of between complexity and the gradient approximation. The larger t is, the more accurate the gradient approximation nut larger the complexity. The authors mentioned that for t "sufficiently large", the approximation would be tight, but is there a way to asses what this value is, and how it affects the complexity in practice?

Q4: Thank you for clarifying this point. I think this should be added in the main paper. Is there a way to choose (ϵ, δ) in a principled way that guarantees that edges are not deleted?

Q6: The reviewers have not clarified this point in my original review.

➔ *Replying to Response to rebuttal*

Official Comment by Authors

Official Comment

by Authors ( Haoyun Zhou (/profile?id=~Haoyun_Zhou1), Honglian Wang (/profile?id=~Honglian_Wang1), Aristides Gionis (/profile?id=~Aristides_Gionis1))

 11 Dec 2024, 07:42

 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors

Comment:

Q1 and Q2: We thank the reviewer for the constructive comments. We agree that the current normalization step differs from the commonly used projection-to-simplex method. Although we want to stress that, it is a widely adopted approach to map the solution to the feasible region and has performed well in our practical experiments. In the next version of the paper, we will revise this projection step and include a convergence analysis to further enhance the work.

Q3: Thank you for your question! In the paper we use the Neumann series to calculate the matrix inverse: $(I - X)^{-1} = \sum_{t=0}^{\infty} X^t$. To answer your question, we need to calculate the error of truncating the series.

Let S_k denote the partial sum of the Neumann series up to the k -th term:

$$S_k = \sum_{t=0}^k X^t = I + X + X^2 + \dots + X^k.$$

The truncation error E_k is defined as: $E_k = (I - X)^{-1} - S_k$.

We can express E_k as the sum of the remaining terms: $E_k = \sum_{t=k+1}^{\infty} X^t$.

Factoring out X^{k+1} from the series we get: $E_k = X^{k+1} \sum_{t=0}^{\infty} X^t$.

Using the formula for the Neumann series, the infinite sum $\sum_{t=0}^{\infty} X^t$ is equal to $(I - X)^{-1}$. Thus: $E_k = X^{k+1}(I - X)^{-1}$.

Taking the matrix norm of both sides, we get: $|E_k| \leq |X^{k+1}| \cdot |(I - X)^{-1}|$.

Given that $|X| < 1$, we have: $|E_k| \leq |X|^{k+1} \cdot |(I - X)^{-1}|$, we conclude that the relative approximation error is given by $|X|^{k+1}$.

In our paper, $X = (1 - \gamma)P$, where, following convention, we set $\gamma = 0.15$. Consequently, $|(1 - \gamma)P|_{\infty} = 0.85$. We conclude that the relative error of using the first t term approximation is 0.85^{t+1} . By setting $t = 50$ the relative error is smaller than 0.0003.

Q4: In general, δ and ϵ are both small. To ensure no edge deletions, we must consider the smallest edge weight in the graph and set the absolute change ϵ accordingly. Specifically, we require $(1 - \delta) \min_{i \neq j} P[i, j] - \epsilon > 0$. Once a δ value (for example 0.1) is chosen, we can calculate ϵ accordingly.

Original question 5 and 6:

We apologize for missing these two questions.

For Question 5: Please kindly refer to the answers provided for W4 to Reviewer 8Rry.

For Question 6: LFPRU and LFPRN can achieve perfect fairness only when both the jump vector and the matrix edge weights are adjusted. In our experiments, we report the fairness loss for LFPRU and LFPRN based on a uniform restart vector and the revised transition matrix, rather than the revised jump vector. This is why they do not achieve zero loss. We thank the reviewer for raising this point, and we will explicitly clarify this in the next version of the paper.

We thank the reviewer again for the valuable comments and discussion!



➔ *Replying to Official Comment by Authors*

response

Official Comment by Reviewer Xfax 📅 11 Dec 2024, 12:16

👁 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors

Comment:

Thank you for your responses. I am a bit conflicted overall since the authors have answered points (3), (4), and (6) to my satisfaction, whereas that to (5) is mostly satisfactory.

What still sticks out like a sore thumb though is points (1)-(2). I find it difficult to comprehend a projected gradient descent algorithm which doesn't actually compute projections (in the proper sense as I mentioned before), nor guarantee descent. Granted, the algorithm seems to work in practice on the chosen datasets, but this relies on post-hoc justification.

Hence, I will keep my initial scores and let the area chairs make the final call.

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